



Application Resilience Circuit Breaker & Rate Limiter

Protecting BOU application and downstream billers
from cascading failures



THE PROBLEM

Why Do We Need Circuit Breaker?



We depend on billers

Our application relies on external Biller systems to fetch bills and process payments.



Billers slow down or fail

If a Biller's servers go down or become extremely slow, Bou Service can freeze up waiting for them.



Traffic can spike suddenly

A sudden massive spike in NPCI traffic could overwhelm and crash the Biller's systems.

How the Circuit Breaker Works



Based on a series of pre-configured metrics, the circuit breaker cuts the connection to a failing Biller. For this purpose, three states are defined in a circuit breaker.

CLOSED Normal

Requests flow freely to the Biller.



OPEN Tripped

If 50% (**configurable**) of consecutive requests fail in a Fetch or Payment, the breaker activates. The application stops sending requests to Biller and immediately responds to NPCI with a “Biller Unavailable” message.



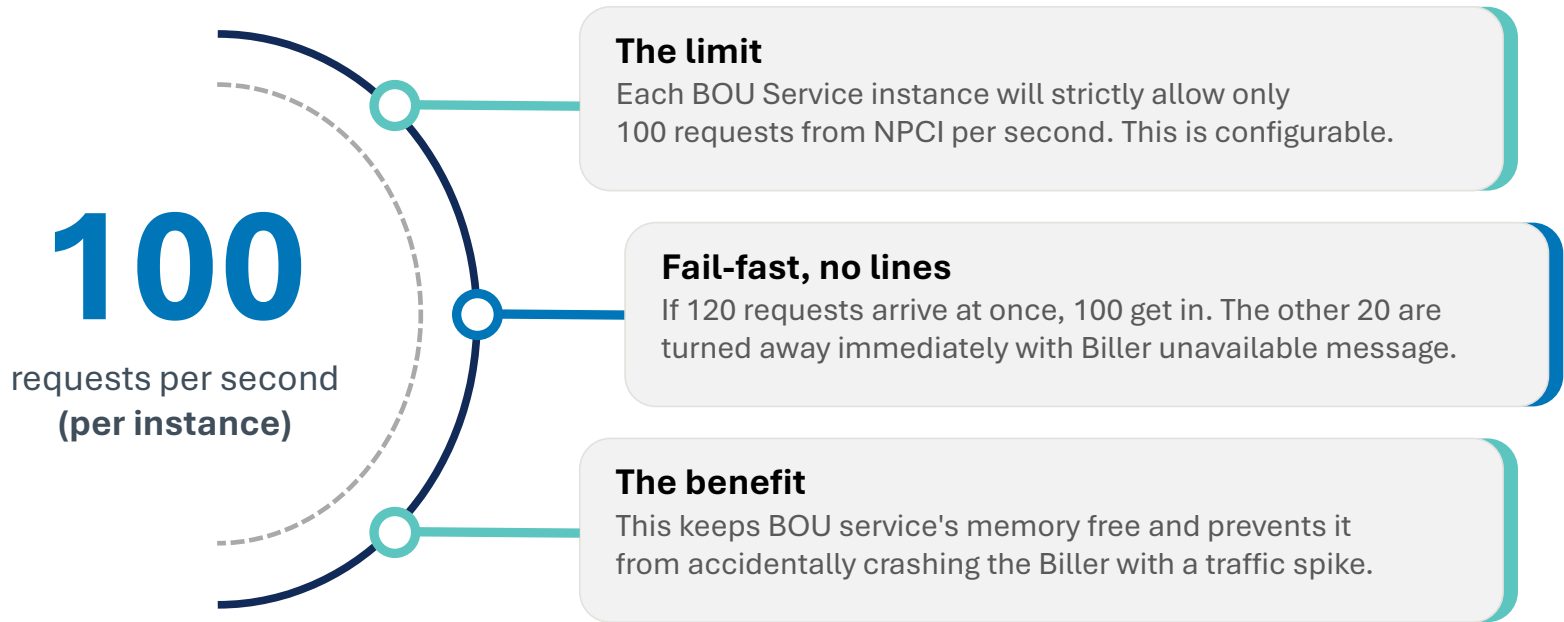
HALF-OPEN Testing

After 1 minute (configurable), it lets 3 test requests through. If they succeed, it returns to normal. If they fail, it activates again.



How the Rate Limiter Works

The Rate Limiter only lets a fixed number of requests through each second.



Layered Security in Action

What happens, in order, every time NPCI sends a Fetch or Payment request for a particular Biller:

1

Circuit Breaker

Checks if the Biller and flow is currently marked as “unavailable” by the Circuit Breaker.



2

Rate Limiter

Checks if NPCI is sending too many requests of the Biller flow this second.



3

HTTP Call

The actual connection made to the Biller.



Isolated by design:

each **Biller** (Biller A, Biller B) and their **flow** (Fetch and Payment) has its own Circuit Breaker, so one broken Biller flow never affects the others.

SUMMARY & BENEFITS

Why This Matters for Our Application



Prevents app freezes

Keeps the application fast and responsive, even when billers are down (due to technical issues, or network problems) or is causing timeouts.



Avoids hammering the Billers

Ensures the application never overwhelms the biller with traffic.



Graceful failures

NPCI gets immediate “Unavailable” message instead of waiting for a dedicated time that will eventually cause extended waiting period to the customer.